

The waveform-first forward-fit reconstruction for the MX17 micro-TPC

from “why doesn’t the track thread the reference?” to a complete reconstruction chain,
a resolved drift-velocity puzzle, a Magboltz-matched gas, and in-situ drift-gap maps

MX17 June cosmic-bench analysis — detectors A (det3), B (det2), E (det4)
primary dataset: det3 Saturday long run, resist 490 V / drift 1000 V, M3 recipe $\chi^2 < 1$, NClus=4

25–27 July 2026

Abstract

We rebuilt the micro-TPC reconstruction from the raw waveforms with no inherited assumptions, to understand why reconstructed charge clusters did not “thread” the M3 reference track through the drift gap. The cause is a general property of resistive-strip readouts: *every* per-strip time estimate is extracted from an aggregate waveform that mixes the strip’s own charge with delayed, dispersed copies of its neighbours’ charge, compressing the time–position ladder by 20–30% and making tracks look $\sim 40\text{--}50\%$ too steep. The fix is a *forward model*: fit the full waveform matrix with a track line, a non-negative charge-depth profile, a measured pulse template and a calibrated sharing kernel. On identical events this reaches a per-event angular residual of $\sigma = 1.0\text{--}1.1^\circ$ versus M3 (production ladder: $5\text{--}6^\circ$; previous best hybrid: $1.5\text{--}1.6^\circ$), threads $\sim 70\%$ of tracks to $< 1\text{ mm}$ over the full gap, and a toy closure shows the remaining $\sim 1^\circ$ is the diffusion/charge-granularity *physics floor* — electronics noise contributes only 0.02° . The method resolves the long-standing drift-velocity tension ($v = 36.7 \pm 0.3 \pm 0.9\ \mu\text{m/ns}$ at 1000 V; the geometric 34.3 was biased by its spatial-floor subtraction), matches Magboltz for Ar/iC₄H₁₀ 95/5 + 0.8% H₂O across the full drift scan, and turns the deconvolved charge-arrival profile into an *in-situ drift-gap map*: det3’s cathode plane is tilted by $\sim 2.7\text{ mm}$, det4’s is bowed by $\pm 3\text{ mm}$, and det2’s is flat at the full 30 mm gap. The chain is packaged as a per-detector calibration protocol plus a production fitter (`wft_reco.py`), validated on three chambers.

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1 Introduction: the threading problem

The MX17 chambers are read out as **micro-TPCs**: a muon crossing the $\sim 30\text{ mm}$ drift gap leaves a line of ionization; the electrons drift to the micromegas mesh at constant velocity, so *arrival time*

encodes depth. Each strip therefore measures a point (position, time), and a straight muon should appear as a straight “ladder” in the position–time plane (Fig. 1).

It did not. With hit-level reconstruction, the fitted ladders were systematically $\sim 4^\circ$ steeper than the M3 reference telescope tracks: in 3-D event displays the reference threads the charge cloud at the mesh and fans away from it with depth (REFERENCE_TRACK_THREADING_REPORT.md). Alignment was ruled out (mesh residuals centred to $< 10\ \mu\text{m}$); hit-level “unsharing” (deconvolution) fixed the bias only on ensemble average while adding per-event noise. The question this work started from: *can any reconstruction make a single muon thread the reference line through the whole gap?*

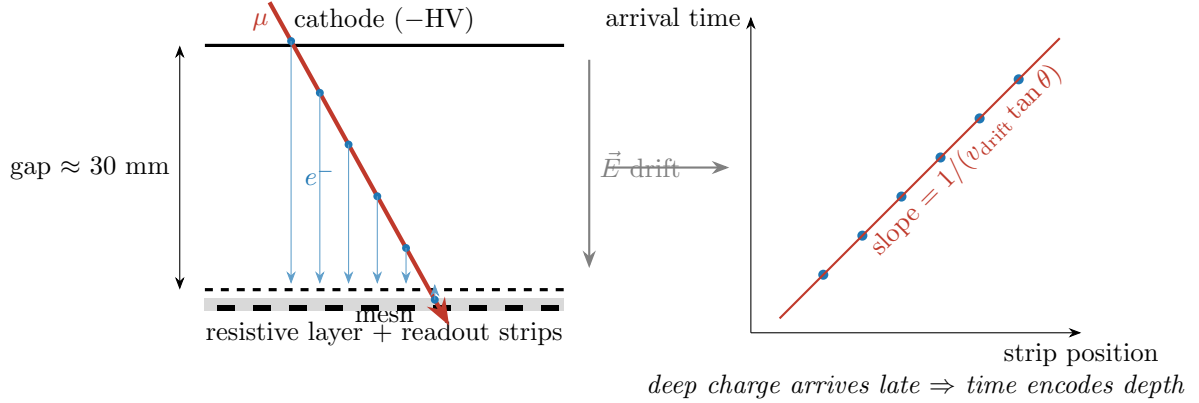


Figure 1: The micro-TPC principle. Left: a muon ionizes along its path; electrons drift at constant v_{drift} to the mesh, are amplified, and induce signals on the strips below. Right: each strip’s charge-arrival time maps its depth, so a straight track is a straight line in the position–time plane. The reconstruction’s job is to measure that line.

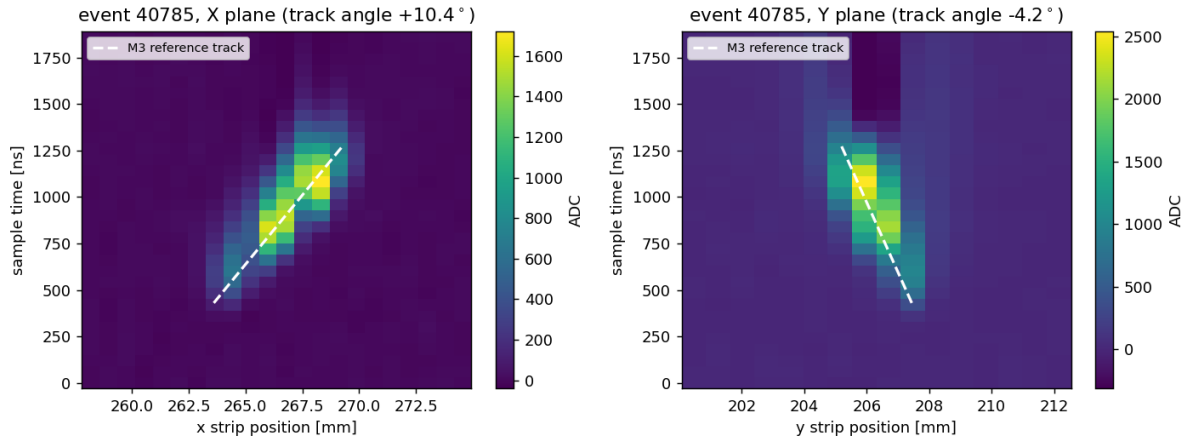


Figure 2: What the raw data looks like. One cosmic muon in the det3 waveform matrix (colour = pedestal/common-noise-subtracted ADC, one column per 0.78 mm strip, one row per 60 ns sample). The inclined charge ridge is the track; the dashed line is the M3 telescope prediction (drawn at $v = 36.6\ \mu\text{m}/\text{ns}$). Note the $\sim 430\ \text{ns}$ trigger latency before the first charge and the dark undershoot band after the pulses.

2 Why hit-based timing fails here: the sharing mixture

The resistive layer deliberately spreads each avalanche’s signal to neighbouring strips. The copy that a neighbour receives is *delayed and dispersed* (measured below: $\sim 29\%$ per first neighbour, delay $\tau_s \approx 47\ \text{ns}$, dispersion $\sigma_s \approx 90\ \text{ns}$). Every strip waveform is therefore a *mixture* (Fig. 3): its own charge at its own arrival time, plus neighbour charge at the neighbours’ (different) arrival times.

the muon crosses strips at different depths
 $\Rightarrow$ different arrival times

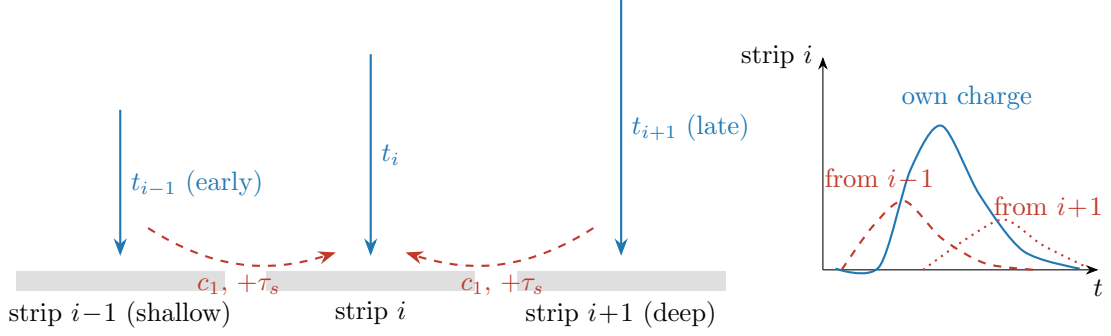


Figure 3: The mixing problem. Strip i 's waveform (right) is the sum of its own charge (solid) and delayed, dispersed copies of both neighbours' charge (dashed/dotted) — which arrived at *different* times because the track crosses each strip at a different depth. Any single number extracted from this aggregate (threshold, CFD, matched filter...) is pulled toward the cluster-mean time.

The consequence is measured in Fig. 4: per-strip times (any estimator) are too *late* near the mesh end of the track and too *early* at the deep end — an S-shaped compression of the ladder by 20–30%. Read with the physical drift velocity, a compressed ladder is a track that is too steep in space; this is the entire “fans away with depth” effect. Because the distortion lives in the aggregate waveform itself, *no better per-strip time estimator can fix it*. It must be modelled.

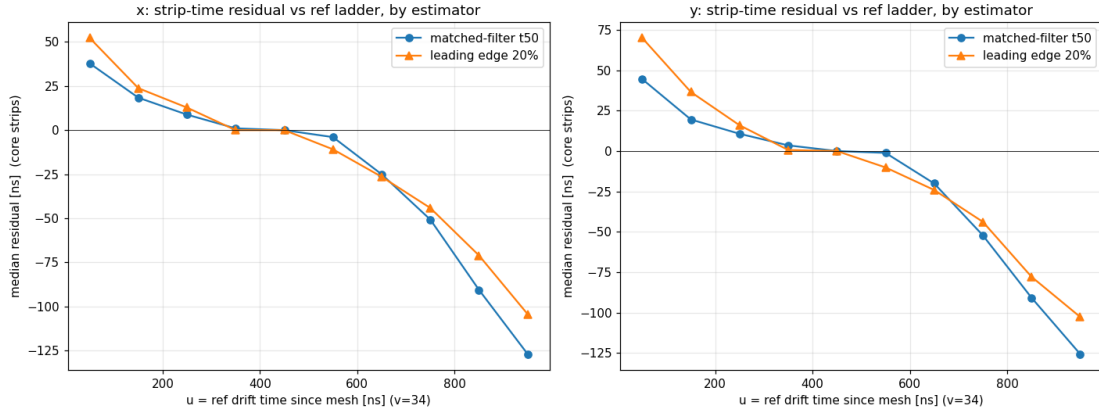


Figure 4: Estimator-independent ladder compression. Median per-strip time residual versus the reference-implied arrival time (per-event t_0 floated), for two very different time estimators. Both are $\sim +40$ ns late at the mesh and -100 ns early at the cathode.

3 The forward model

Instead of extracting times and fitting points, we *predict the entire waveform matrix* from a physical track hypothesis and fit the raw samples (Fig. 5). Sharing is applied *forward* in the model — never inverted — which is numerically well-conditioned where deconvolution is not.

Parameters (per plane). A track line in position–time space,

$$\text{pos}(u) = p_0 + w u, \quad w = v_{\text{drift}} \tan \theta \text{ [mm/ns]},$$

with u the drift time since the mesh crossing at t_0 ; plus a **non-negative charge profile** q_k in K bins of 60 ns of arrival time (the amount of ionization arriving in each depth slice — left free

because ionization fluctuates). Note v_{drift} never enters the fit itself: w is measured in mm/ns directly.

Model. The predicted waveform of strip i at sample time t :

$$m_i(t) = \sum_k q_k \left[F_{ik} h(t - t_0 - u_k) + c_1(F_{i-1,k} + F_{i+1,k}) h_s(t - t_0 - u_k - \tau_s) + c_2(\dots 2\tau_s) \right]$$

where F_{ik} is the geometric fraction of bin- k charge landing on strip i (track-segment length $\otimes$ Gaussian spread $\sigma_p(u) = \sqrt{\sigma_{p0}^2 + D_p^2 u}$), h is the *measured* impulse-response template (Fig. 6), and h_s is the same template dispersed by σ_s — the resistive sharing kernel ($c_1, c_2, \tau_s, \sigma_s$).

Fitting. For fixed (p_0, w, t_0) the model is *linear* in the charges q_k , solved by non-negative least squares in milliseconds; the three track parameters are found by a coarse grid plus a small simplex (`forward_model3.py`, ~ 0.1 s/plane, 800 events in 42 s on 14 cores). Saturated samples are censored (one-sided penalty). Per-channel gains (measured spread only 1.4%) are divided out.

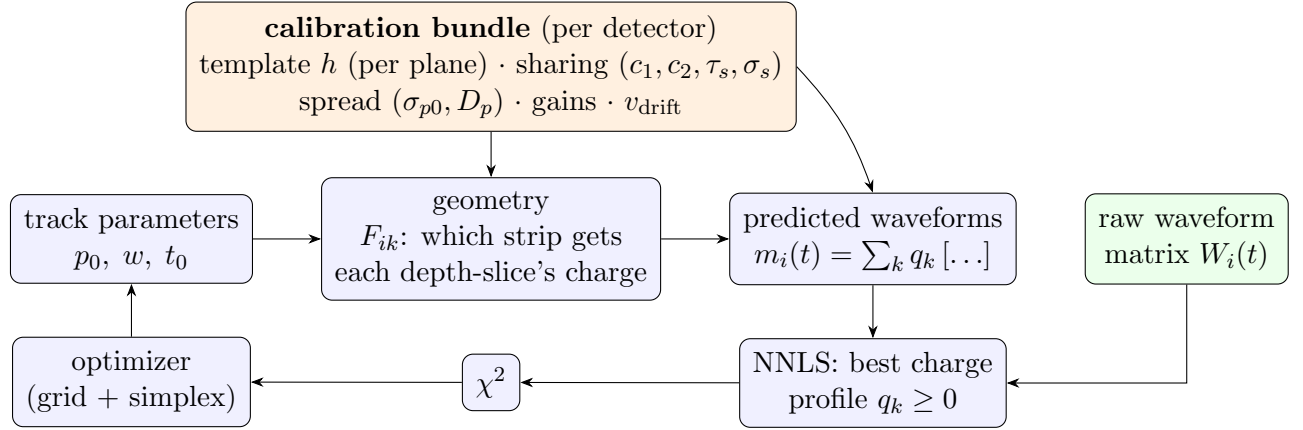


Figure 5: The fit loop. The optimizer proposes a track line; the model predicts every sample of every strip (charge profile solved linearly, with positivity); the χ^2 against the raw data drives the next proposal. Output: track position, angle, charge-vs-depth profile, and honest per-event errors.

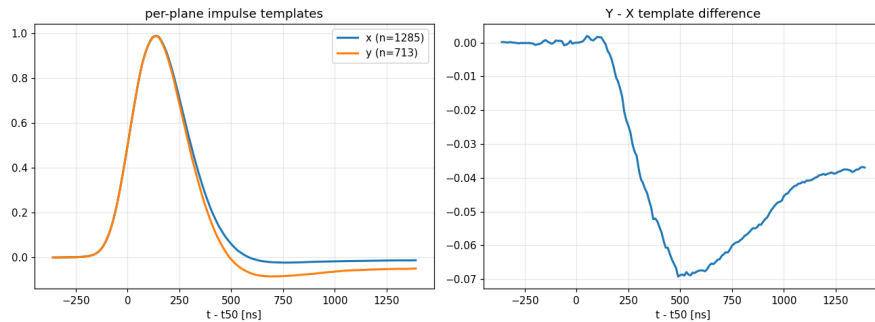


Figure 6: The measured impulse template (median of ~ 2000 bright strips on steep tracks, where one strip's charge arrives nearly at once). X and Y have identical rise (180 ns 10–90) and width, but Y's undershoot is 4 $\times$ deeper — one reason the planes must be calibrated separately.

Calibration protocol (per detector, automated).

1. **Template:** median waveform of bright strips on steep tracks, per plane.
2. **Gain map:** per-channel data/model amplitude ratio (det3: 1.4% spread).

3. **Hyper-fit:** with the track line *pinned to the M3 reference* on ~ 150 medium-angle muons, fit $(c_1, c_2, k_Y, \tau_s, \sigma_s, \sigma_{p0}, D_p, v_{\text{drift}})$ by minimizing the summed waveform χ^2 . (k_Y scales the Y-plane sharing.)

det3 result: $c_1 = 0.29$, $c_2 = 0.05$, $k_Y = 1.38$, $\tau_s = 47$ ns, $\sigma_s = 90$ ns, $\sigma_{p0} = 0.09$ mm, $D_p = 0.008$, $v = 36.60$ $\mu\text{m}/\text{ns}$. Anti-circularity controls: calibrating instead against the (biased) production ladders converges 49% *worse* in χ^2 with contorted parameters, and free fits still pull toward the reference — the waveforms genuinely prefer the reference-sloped tracks.

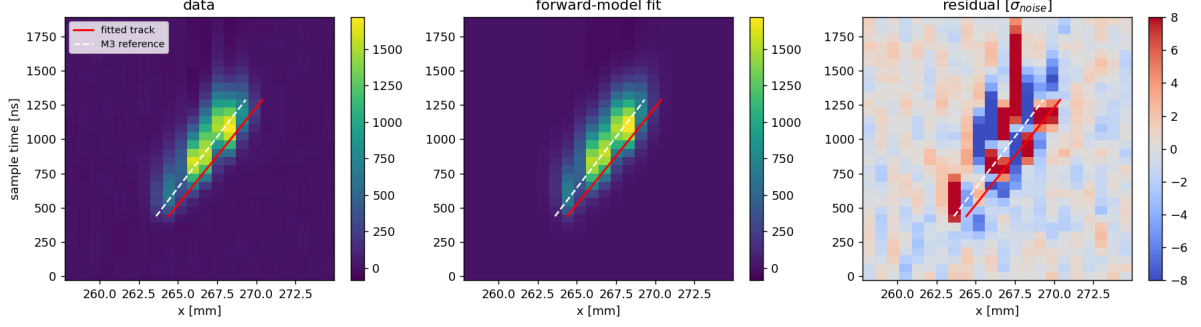


Figure 7: One event, fitted. Data, and normalized residual for the X plane of the event in Fig. 2. Fitted angle 10.9° vs M3 reference 10.4° .

4 Performance

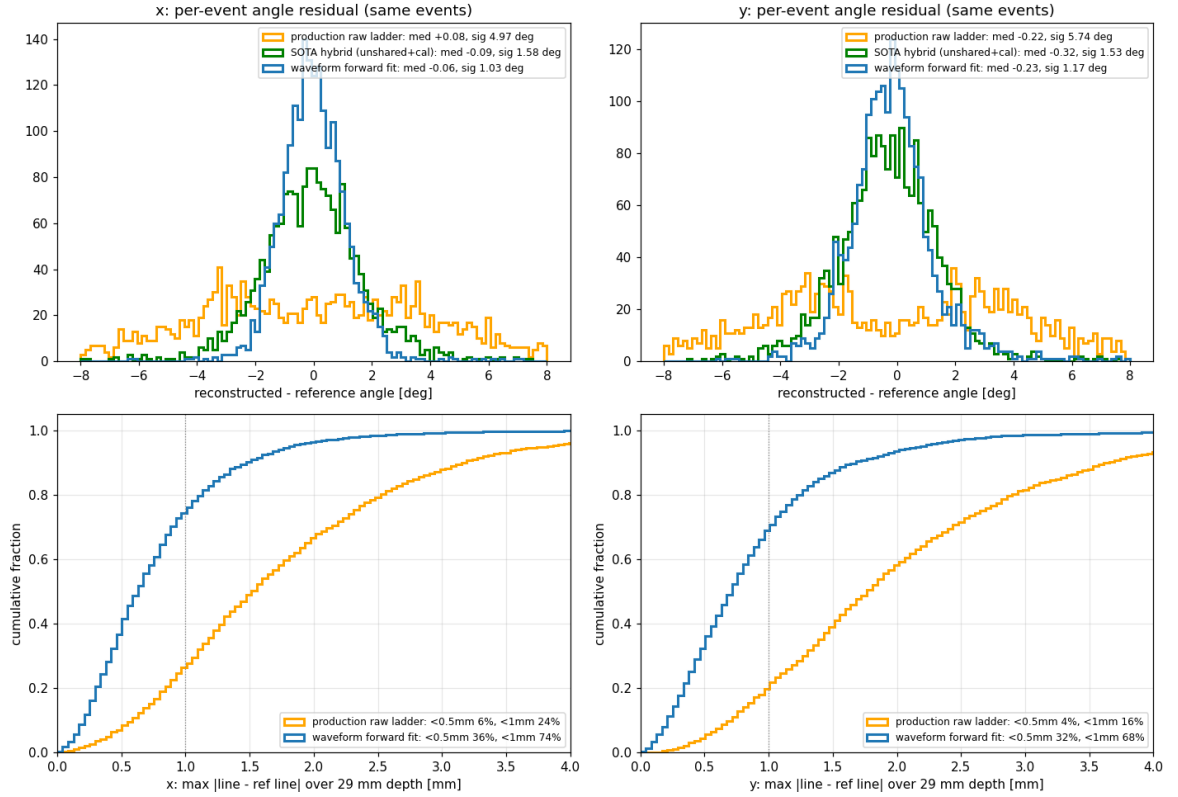


Figure 8: Head-to-head on identical events (2,000-event disjoint test set). Top: per-event angle residual vs M3. Bottom: cumulative worst deviation of the fitted line from the reference line over the full 29 mm depth.

reconstruction	angle σ (x/y)	mesh σ (x/y)	<1 mm full depth	implied- v flat?
production raw ladder	4.97°/5.74°	0.67/0.76 mm	24%/16%	no (56→39)
SOTA hybrid (unshared+cal.)	1.58°/1.53°	—	—	—
matched-filter ladder + remap	1.51°/1.86°	0.79/0.73 mm	54%/52%	no (± 9)
forward fit (v2)	1.06°/1.10°	0.57/0.59 mm	74%/70%	yes (± 1)

Table 1: det3 benchmark, $|\tan \theta| \in [0.10, 0.40]$, same events throughout. The remap row is a cheap production pre-pass (matched-filter times + one trained slope calibration): SOTA-level width but biased vs angle.

Across *all* angles the slope quality is flat at $\sim 1.0\text{--}1.1^\circ$ for $|\tan \theta| > 0.04$ and degrades gracefully to $1.4^\circ(\text{x})/2.4^\circ(\text{y})$ below (timing carries no slope information on 1–2 strips; the production fitter flags these `slope_reliable=False` and can fall back to a joint two-plane fit). Mesh-position quality is *best* at near-vertical (0.47–0.53 mm).

The physics floor. A toy closure test (real geometries, real noise, model-generated waveforms, known tracks) shows the fit recovers angles to 0.02° with electronics noise alone; adding a 0.30 mm RMS transverse jitter to each 60 ns depth-slice’s charge centroid — the expected scale of transverse diffusion sampled by few avalanche-weighted electrons — reproduces the observed 1.05° *exactly*. **The residual per-event scatter is detector physics, not reconstruction.** Component ablations confirm robustness: no single model ingredient shifts the angle width by more than 0.05° ; the essential ingredients are the sharing kernel and the per-plane template.

5 Drift velocity: the tension resolved, the gas identified

The calibration prefers $v = 36.6 \mu\text{m/ns}$ while the previous geometric estimator gave $34.3 \mu\text{m/ns}$. Four independent tests (Fig. 9) locate the discrepancy entirely on the geometric side:

- the geometric method’s *time* ingredient (676 ns) agrees with our deconvolved gap-crossing duration (674 ns); its *extent column* (23.2 mm effective) disagrees with its own time-free measurement (24.5 mm) — a 1.3 mm ambiguity in subtracting the ~ 2 mm resistive-spread floor, worth exactly the 5–7% velocity difference;
- toys calibrated with a deliberately wrong sharing truth at $v_{\text{true}} = 34$ return 33.1: model mismatch *deflates* v , it cannot inflate 34 into 36.6;
- the profile likelihood (sharing re-fit at every v) has a sharp minimum at 36.7 with weak $v \leftrightarrow \text{sharing}$ degeneracy;
- the fitted transverse speed divided by the reference tangent is flat vs angle (physical), where the raw ladder’s falls 56→39 (unphysical).

Result: $v(1000 \text{ V}) = 36.7 \pm 0.3 \text{ (fit)} \pm 0.9 \text{ (model)} \mu\text{m/ns}$.

Ref-pinned $\chi^2(v)$ scans on every drift-scan point give a smooth $v(E)$ from 12 to $39 \mu\text{m/ns}$ (300–1100 V). Against a 71-mixture Magboltz library (including 41 new tables computed this session, locally and on lxplus condor, cross-validated to $0.01 \mu\text{m/ns}$), the curve lands on **Ar/iC₄H₁₀ 95/5** + (0.81–0.86)% **H₂O** at RMS 0.63–0.67 $\mu\text{m/ns}$ for the operating fields (Fig. 10) — consistent with the known bottle-drying trajectory that week. The 34-vs-36.6 question was, in gas terms, a $\sim 0.5\%$ humidity ambiguity.

6 The drift-gap maps: an apparent 25 mm gap that wasn’t

The fitted charge profile q_k is the template- and sharing-deconvolved *charge-arrival distribution*; for a through-going muon its endpoint U is the full-gap drift duration, so $v \cdot U$ measures the local

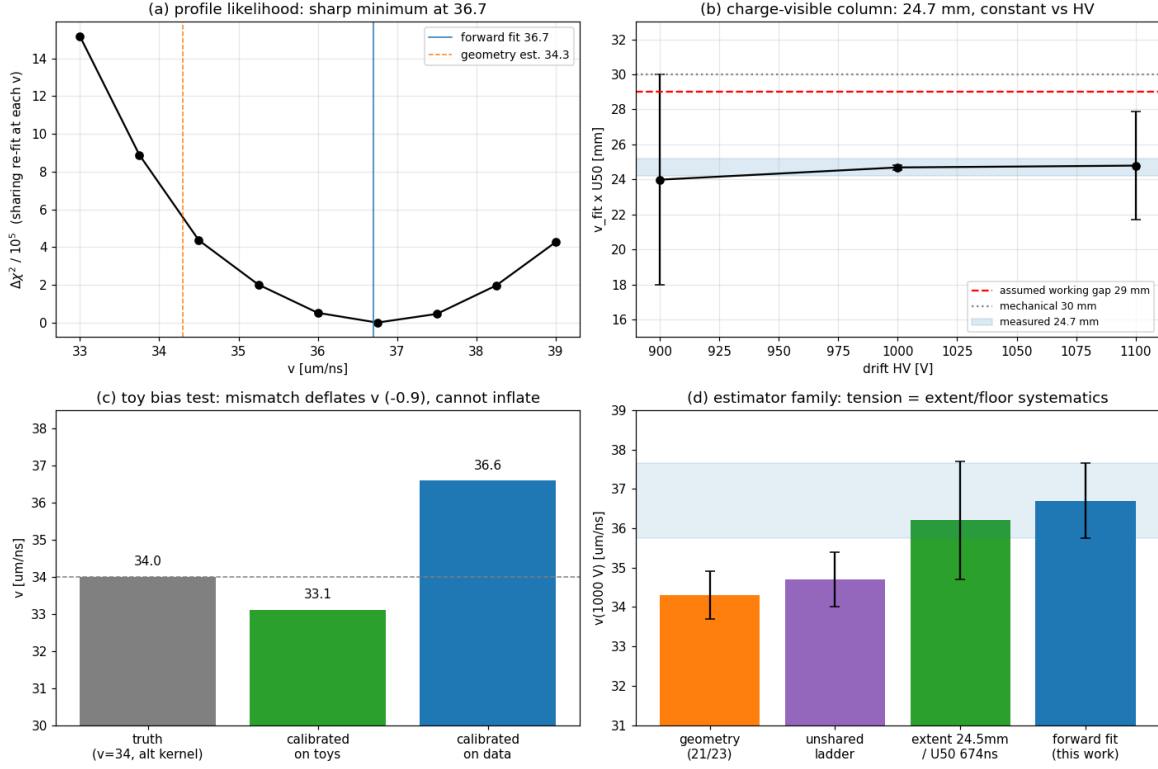


Figure 9: The velocity-tension resolution. (a) profiled $\chi^2(v)$; (b) the charge-visible column vs HV (see §6 for the final interpretation); (c) toy bias test; (d) the estimator family.

gap. A first pass gave an alarming 24.7 mm. Overnight adversarial review found and fixed two estimator biases (a median over *sparse* NNLS profiles, and free-fit rather than reference-pinned profiles); the corrected, closure-validated measurement is $U_{50} = 762 \text{ ns} \Rightarrow \mathbf{27.9 \text{ mm}}$, stable against the charge-basis length (Fig. 11), with toys recovering known step-columns to $\leq 5 \text{ ns}$ even under deliberate template mismatch.

Crucially, the data edge is *soft* where toy step-gaps come back sharp: the endpoint **varies with position**. Mapping U_{50} versus the track’s location (Fig. 12) shows det3’s column running 26.5→29.5 mm — a $\sim 2.7 \text{ mm}$ **tilt of the cathode plane along y** plus a slight bow in x — with a small additional electrostatic pull at higher drift field (28.4→27.8 mm from 900→1100 V, the expected E^2 trend). det4 shows its own $\pm 3 \text{ mm}$ bow (28.8 mm average). The control: **det2 (detector B) is flat at 30.5 mm** — the method reads the full mechanical gap on a flat chamber.

Two corollaries: (i) gap-based velocity estimators inherited a -3 to -7% bias from assuming 29 mm everywhere — part of the historical velocity confusion; (ii) the flat deconvolved profile shows the old amplitude-vs-depth “attenuation” is mostly diffusion + per-strip threshold, not charge loss. The $U_{50}(x, y)$ map should simply become part of each chamber’s calibration bundle — it is a free, in-situ flatness survey.

7 The production reconstruction stack

The deliverables, all in `mx_june_cosmic_qa/waveform_first_threading/`:

- `wft_reco.py` — production API. Loads a calibration bundle; `fit_plane()/fit_event()` return position, angle, charge profile, χ^2 , errors (statistical curvature $\oplus$ the $1^\circ/0.33 \text{ mm}$ physics floor), `slope_reliable` and `quality_ok` flags, and a joint two-plane fallback for near-vertical planes.

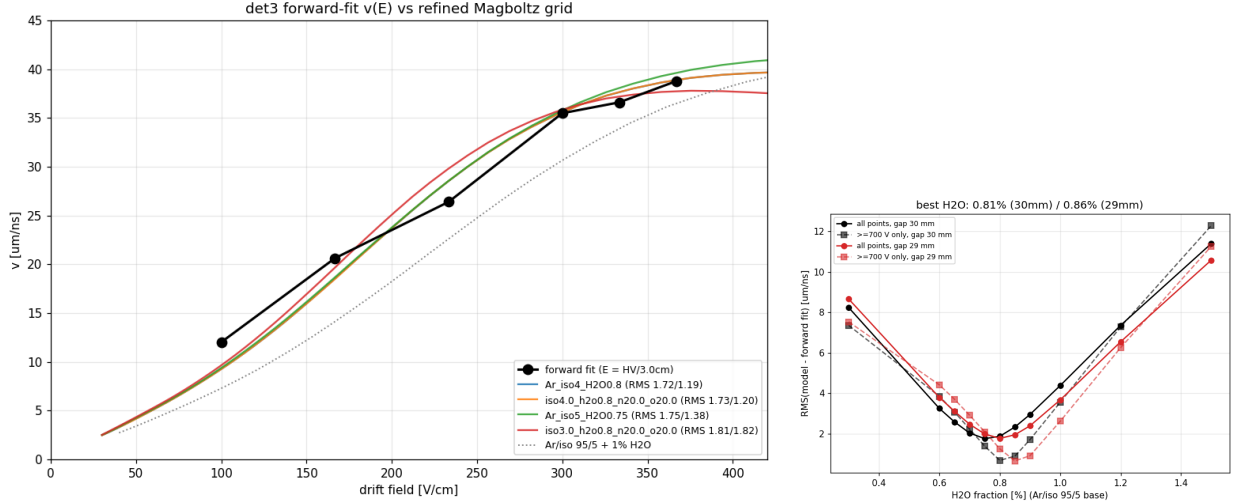


Figure 10: The gas. Left: forward-fit $v(E)$ against the best Magboltz mixtures. Right: RMS vs water fraction — a clean parabola with minimum at 0.81–0.86% H_2O . (The 300/500 V measured points carry an extra charge-basis caveat and should be re-scanned before quantitative reuse; the ≥ 700 V conclusion is unaffected.)

	det3 (A)	det2 (B)	det4 (E)
kernel $c_1 / k_Y / \tau_s / \sigma_s$	0.29 / 1.38 / 47 / 90 ns	0.29 / 1.42 / 49 / 91 ns	0.24 / 2.36 / 68 / 234 ns
v_{drift} (run conditions)	36.6 $\mu\text{m}/\text{ns}$	39.9 $\mu\text{m}/\text{ns}$	34.2 $\mu\text{m}/\text{ns}$
angle σ (x/y)	1.06°/1.10°	1.33°/1.54°	2.00°/1.97°
mesh σ (x/y)	0.57/0.59 mm	0.76/0.56 mm	0.93/0.71 mm
column / topography	27.9 mm, y-tilt	30.5 mm, flat	28.8 mm, bowed

Table 2: The calibration protocol and fitter, run end-to-end on three chambers. det2/det3 share a nearly identical sharing kernel (same production batch); det4 is the gain-limited chamber and would profit from a refinement pass.

- **forward_model2/3.py** — the model and the fast fitter ($\sim 5\times$ speedup, finds a lower χ^2 than plain simplex in 20% of events; handles different DAQ window lengths).
- **Calibration scripts** 03/11/12/13 (template, gains, hyper-fit) and the per-chamber pipelines (26/26b/28/36).
- **Physics tools:** ref-pinned $\chi^2(v)$ for gas monitoring (works even at low HV where gap methods fail), the $U_{50}(x, y)$ gap mapper, and the endpoint-closure toys that validate both.
- A cheap **pre-pass** (matched-filter ladder + trained slope remap) reaching SOTA-level width at negligible cost, for monitoring only.

8 Conclusions and next steps

The waveform-first forward fit turns the MX17 micro-TPC into what it was meant to be: a per-event 3-D tracker with $\sim 1^\circ$ angles and ~ 0.6 mm positions, at the physical information limit set by diffusion and charge granularity. Along the way it resolved the drift-velocity puzzle ($36.7 \pm 0.3 \pm 0.9 \mu\text{m}/\text{ns}$, gas = wet 95/5 at 0.8% H_2O), retracted an apparent short drift gap in favour of measured cathode-plane topography, and produced per-chamber calibrations for three detectors.

Open items: (1) confirm the det3 tilt / det4 bow mechanically at the next opening (no disassembly needed on this evidence alone); (2) an independent humidity handle to pin $|v|$ below $\pm 1 \mu\text{m}/\text{ns}$; (3) re-scan the 300/500 V velocities with the extended charge basis; (4) det4 refinement pass and the remaining chambers (det6/7); (5) a numba/C++ port for another $\sim 10\times$ speed; (6)

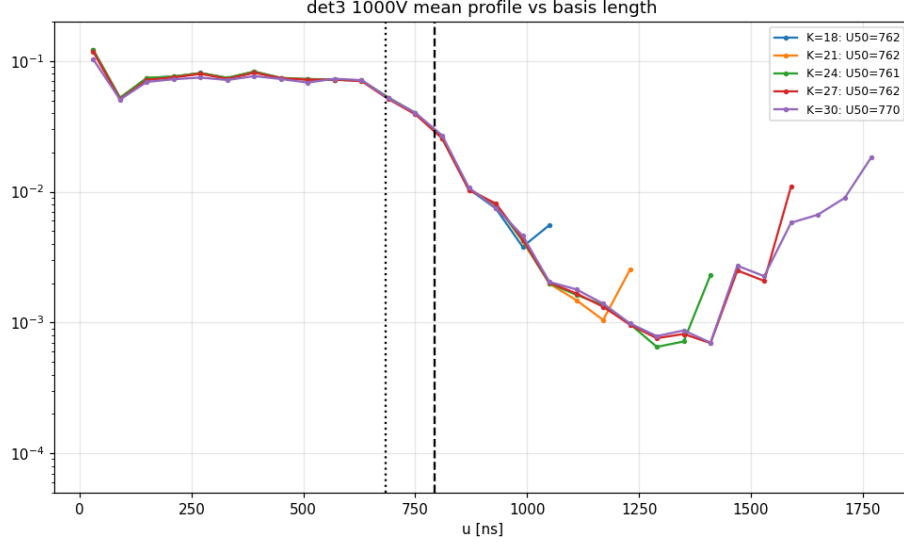


Figure 11: Endpoint robustness. The mean deconvolved charge profile at 1000 V for charge-basis lengths $K = 18 \dots 30$: $U_{50} = 762$ ns for all valid K (junk accumulates only at $K = 30$; closure toys quantify the regime). Dotted/dashed guides: 25 and 29 mm equivalents.

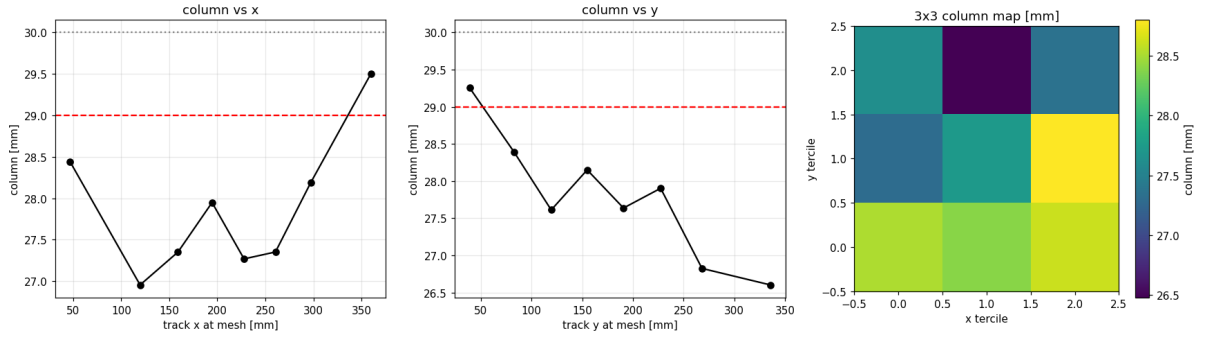


Figure 12: det3's drift-gap topography, measured in situ from cosmic data alone: charge-column length vs track position. The y-tilt of ~ 2.7 mm and slight x-bow are coherent across 7,240 profiles.

apply the chain to the July beam runs of the same chambers.

Reproduction: `WAVEFORM_FIRST_THREADING.md §8/§15` and scripts 01–36 in the package directory. Figures and caches: `<Analysis>/mx17_3/waveform_first/` (and `mx17_2`, `mx17_4` for detectors B/E).

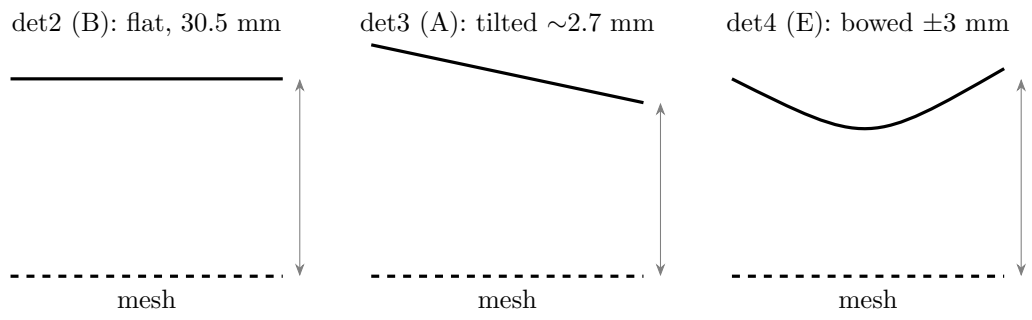


Figure 13: Chamber summary (cartoon). The gap is nominal at the frame on all three chambers; det3's cathode plane is tilted, det4's bowed, det2's flat. Absolute scales carry each chamber's v calibration (± 3 –5%); the patterns are robust.